

A detailed illustration of a steam locomotive, likely a Prussian P8, in a rugged, mountainous landscape. The locomotive is dark with a prominent smokestack and a large boiler. The background shows steep, rocky cliffs and a small body of water. The title 'harzbahn 1873' is overlaid on the image in a stylized, gothic font. The word 'harzbahn' is in white with a black outline, and '1873' is in a dark blue color with a white outline.

harzbahn 1873

RULES OF PLAY

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GAME COMPONENTS

The game consists of the following components:

- 1 game board with a map representing the central Harz Mountains, stock market, and the bank pool
- 7 concessions for railway companies
- 15 certificates for mines
- 13 charters for public companies
 - 8 railway companies
 - 5 mining companies
- 130 share certificates for public companies
- 35 train and machine cards
- 111 track tiles in the colours yellow, green, brown, and grey
- 5 turn order cards
- Wood tokens
 - 1 round marker
 - 1 +50 marker for the Harzer Werke
 - 13 stock value markers (one for each public company)
 - 15 mine markers
 - 54 station markers
 - 30 machine markers
 - 30 switcher markers
- Rules Booklet and Player's Guide
- A bank of approximately 50,000 Marks must be supplied by the players

THE MAP

The game board represents the central area of the Harz Mountains in Germany located between Wernigerode (B9), Halberstadt (B19), Quedlinburg (E20), Nordhausen (J7), and the Brocken Mountain Peak (C4), the renowned highest mountain of Northern Germany (1,141 metres). The map is divided into hexagonal spaces (also called **hexes**) and shows the following features:

- **Topographic Features:** There are hexes with mountains that are marked with the cost to build on that hex. There are also thick black barriers between certain hexes representing impassable terrain.
- **Railway Track:** Some track has already been constructed in some of the hexes. Thick black lines represent track of railway companies that operate more or less outside the game. For game purposes, such track is called **state railway track**. Black and white dashed lines represent existing narrow gauge track that may be used later by the railways of this game. Thin black lines do not represent track; they depict the paths of the **concession routes** that need to be built by the respective railway companies.
- **Towns and Villages:** There are towns and villages represented on the map. All hexes with a purple border contain **towns** that are represented by one or two circles. **Villages** are represented by either a black dot (if there is no track in the hex) or, as in the case of towns, by one or two circles with existing track. Two circles with non-connected tracks are still only one village with two independent stops or stations.
- **Industries:** A total of 20 industry locations are shown on the map, each represented by a circle divided into an upper and a lower sector. 15 of these industries are **mines**, represented by an upper number indicating the mine number (cf. 2.1), and a lower number which is the multiplier used for calculating revenue (cf. 4.4.4). The five remaining industries are **factories**, which have a characterizing abbreviation in their upper sector and a revenue value in their lower sector. The following abbreviations are used:
 - ZW “Zuckerwerk”: sugar refinery
 - SB “Steinbruch”: quarry
 - PM “Papiermühle”: paper mill
 - SM “Sägemühle”: saw mill
 - Hex C6 has only the quarry Knaupsholz, but no village.

1. COURSE OF THE GAME

The course of the game is divided into **trading rounds** (TRs) and **operating rounds** (ORs).

- In the TRs, players invest in mining and railway companies (generally called **companies**).
- In the ORs they act for companies as directors.

The game starts with a trading round.

After each trading round there is a set of one to three operating rounds, depending on the **game phase** (cf. Rule 5, Game Phases).

Each TR is further divided into an **auction round** (AR) followed by a **stock round** (SR).

*Before continuing with the rules, new players should first read the **General Game Description** in the Player's Guide, where a more detailed overview of the course of the game is provided.*

2. GAME START

Place the game board on the table. Now get all the components ready that are required at the start of the game:

- Gather the 15 mine certificates, the concessions and charters of the HBE and the GHE, the HW charter, and the yellow track tiles.
- Place the ten MHE shares in the bank pool.
- Sort the train/machine cards into a pile with the 1-card on top, followed by the 2-cards, and so on. Put them into the bank pool. Separate the seven diesel railcar cards and place them into the bank pool in a separate pile.
- Place the 15 mine markers on the map in their respective locations with their white active sides face up.

Next, randomly deal out the turn order cards to determine the **start turn order**. The first player in this order is called the **start player**. The **start player** assumes also the management of the **bank**. The bank possesses everything that is not owned by players or companies (money, mines, concessions, shares, and trains/machines). The assets of the bank must be kept strictly separate from players' and companies' assets.

Now, every player gets their **start capital** according to the following figures.

- 2 players: 2,100 Marks per player
- 3 players: 1,400 Marks per player
- 4 players: 1,050 Marks per player
- 5 players: 840 Marks per player

The bank has unlimited money for game purposes. If the situation occurs that there is not enough money in the bank, each player should return the same amount of money to the bank so as to provide sufficient cash assets to keep the bank liquid for the duration of the current round. Keep track of any returned cash, since it is still part of the players' assets. During the last set of ORs, revenue for the players can be simply written down to speed play.

2.1 START AUCTION ROUND

After each player has received their start capital, the game commences with an initial auction round (the **Start Auction Round**), in which only the HBE and GHE railway concessions and the 15 mines can be acquired. A table of the mines and concessions is shown on the next page.

The mines and the two concessions are auctioned as follows. This auction is round robin and carried out in two parts.

During the first part, players bid a **start premium** for the right to purchase the first item. On their turn, they must either bid or pass. The minimum bid is zero and all higher bids must be multiples of ten. Players that pass are out of this part of the auction.

Once all players have passed, turn order is again determined, with the winner of this part of the auction going first and the others following in reverse order in which they have passed.

Now the second part of the **Start Auction** commences. At first, the player that won the first auction part must buy one of the available mines or concessions, paying its face value (mines) or minimum price (concessions) plus the start premium (equal to the highest bid in the auction's first part) to the bank.

Then the auction proceeds in the new turn order:

- On a player's turn, they must pass, buy *one* mine, or buy *one* of the two available concessions from the bank. At first, players must pay the start premium in addition to the minimum price when they buy one of the available items.
- Once all players have passed successively, the premium is reduced by 10 Marks. This reduction takes place each time all players passed in succession.
- The chance to buy always goes in turn order throughout the whole auction.
- The Start Auction Round ends when all offered mines and concessions have been purchased by the players. It also ends when all players pass successively after the premium has dropped to 0 Marks.
- All unpurchased mines are now closed and their certificates are returned to the bank. Turn their markers on the map to their red "closed" sides. Closed mines can only be reopened when a public mining company buys them (cf. 4.3.4). They can no longer be acquired by players. Concessions that are not bought during the Start Auction will again be on offer in the next AR.

The Player's Guide has a complete commented example of a Start Auction.

2.2 PROGRESS OF THE GAME AFTER THE START AUCTION ROUND

The Start Auction Round is followed by a normal stock round (cf. 3.2), retaining the turn order from the second part of the Start Auction.

*Clarification: Once purchased by players, and as long as they possess them, mines are considered to be small independent mining companies (**independent mines**).*

TABLE 1. AVAILABLE MINES AND CONCESSIONS

# (HEX)	LOCATION	FACE VALUE	BASE REVENUE	MAXIMUM REVENUE	REMARKS
1 (E8)	Königshütte	110	40	140	Vor-Harzer mine
2 (E4)	Wurmberg	120	40	170	--
3 (I16)	Silberhütte	130	40	170	--
4 (D11)	Hüttenrode	140	40	170	Vor-Harzer mine
5 (D13)	Braunesumpf	150	50	160	Vor-Harzer mine
6 (E10)	Rübeland	160	50	190	Vor-Harzer mine
7 (I14)	Lindenberg	170	50	220	--
8 (I8)	Netzkater (Rabensteiner Stollen)	180	60	210	--
9 (G2)	Wieda	190	60	240	--
10 (D9)	Elbingerode	200	60	240	Vor-Harzer mine
11 (F7)	Tanne	220	70	230	Vor-Harzer mine
12 (D15)	Blankenburg	240	70	230	Vor-Harzer mine; Harzer Werke AG starting privilege
13 (I18)	Harzgerode	260	70	260	--
14 (G4)	Zorge	280	90	270	Vor-Harzer mine
15 (F15)	Thale	300	90	300	Starts connected to the state railway
HBE concession	--	100 (min. price)	--	--	Concession
GHE concession	--	100 (min. price)	--	--	Concession

TABLE 2. RAILWAY COMPANIES REPRESENTED IN THE GAME

RAILWAY COMPANY	HOME STATIONS	CONCESSION IS ACTIVATED IN GAME PHASE	STATION MARKERS FOR 20% / 10% SHARES	CONCESSION ROUTE AND ITS TOTAL CONSTRUCTION COST
Halberstadt- Blankenburger Eisenbahn (HBE)	Halberstadt (B19) Blankenburg (D15)	1	8 / 12	Halberstadt- Blankenburg cost: 0 M
Gernrode-Harzgeroder Eisenbahn (GHE)	Gernrode (G20) Harzgerode (I18)	1	6 / 9	Gernrode-Harzgerode cost: 150 M
Nordhausen- Wernigeroder Eisenbahn (NWE)	Nordhausen (J7) Wernigerode (B9) Benneckenstein (G6)	3 (2)*	6 / 9	Nordhausen- Wernigerode cost: 500 M
Südharzeisenbahn (SHE)	Walkenried (I2) Braunlage (E4)	3 (2)*	4 / 6	Walkenried-Braunlage cost: 300 M
Kleinbahn Ellrich-Zorge (KEZ)	Ellrich (I4) Zorge (G4)	3 (2)*	4 / 6	Ellrich-Zorge cost: 100 M
Wernigerode- Blankenburger Eisenbahn (WBE)	Wernigerode (B9) Blankenburg (D15)	4 (3)*	4 / 6	Wernigerode- Blankenburg cost: 0 M
Quedlinburger Lokalbahn (QLB)	Quedlinburg (E20)	4	4 / 6	no concession route
Magdeburg- Halberstädter Eisenbahn (MHE)	–	–	–	external company; see 3.2.7 and 4.5

*Note: The concessions of the KEZ, the SHE and the NWE are normally activated in Game Phase 3. However, they become active earlier, in Game Phase 2, as soon as a tile is laid in one of their respective concession route hexes (by another railway company or an independent mine). In the same way, the concession of the WBE, normally activated in Game Phase 4, becomes active in Game Phase 3 as soon as a tile is laid in one of its concession route hexes (except the Blankenburg hex; see Rule 4.6).

3. TRADING ROUNDS (TRs)

At the beginning of the game and between any two sets of operating rounds there is a trading round (TR). Each trading round consists of an auction round (cf. 3.1) followed by a stock round (cf. 3.2).

3.1 AUCTION ROUNDS (ARs)

The first auction round is called the Start Auction Round (cf. 2.1). In later ARs, unsold active concessions are available for auction. At the beginning of the game, only the concessions for the HBE and the GHE are active. The concessions for the other railway companies are activated by certain events in the course of the game (shown in the table on the preceding page). Also, a "purchase option" to buy shares of a company in receivership may be available for auction (cf. 6.3, Release from Receivership).

These later ARs differ from the Start Auction Round and are carried out as follows:

- In the current turn order (cf. 3.2.8) the players have a round robin opportunity to auction an active concession or, if there are companies in receivership, a purchase option for shares in such a company.
- During a player's turn, they must either pass or auction off an available concession or purchase option. The player designates the concession or purchase option they would auction, and gives a starting bid for it. This starting bid must be at least 100 Marks and a multiple of 10 Marks.
- In turn order, the players now have the opportunity to submit higher bids. The minimum bid increase is 10 Marks, and each bid must always be a multiple of 10 Marks.
- Once a player has passed, they cannot bid further on this concession or purchase option during the current auction.
- When all players have passed successively, the player with the highest bid gets the concession or purchase option and pays the bid amount to the bank.
- If there is still a concession or purchase option available, the next player has now the opportunity to auction off one of them.
- This continues until there is nothing left to be auctioned or all players have successively passed on the opportunity to auction an available concession or purchase option. This ends the auction round.
- Unauctioned active concessions or purchase options remain available to auction in the next auction round.
- In many ARs there is nothing at all to be auctioned. In such a case, the auction round ends immediately.
- After the end of an auction round, a stock round (the second part of the current TR) follows immediately.

3.2 STOCK ROUNDS (SRs)

For the first stock round, the players use the turn order of the second part of the Start Auction. For all other stock rounds, turn order is re-determined according to 3.2.8.

During a player's turn, they first have the opportunity (except in the first SR) to sell one or more shares (cf. 3.2.2). After that, they have the opportunity to carry out at most one of the following actions:

- Purchase of a share (cf. 3.2.3)
- Form a railway company (cf. 3.2.4)
- Form a public mining company (cf. 3.2.5)
- Exercise of a purchase option (cf. 3.2.6)

If the player does not wish to perform an action, they pass, and the next player's turn commences.

3.2.1 PUBLIC COMPANIES: SHARES, STOCK VALUES, AND DIRECTORS

Once a public company is formed, it has company **shares**. These shares can be:

- Two 50% shares (public mining companies only)
- Five 20% shares
- Ten 10% shares

The current value of a share in a company is always the **stock value** of this company on the **stock market** regardless of whether it is a 50%, 20%, or 10% share.

Once formed, each company that is not in receivership has a **director** and the player who forms the company is always its first director. Later, the directorship can change. This happens as soon as another player has *more* shares in a company than the previous director. Change of directorship can happen during a sale (cf. 3.2.2) or a purchase (cf. 3.2.3).

Exception: The directorship of a railway company remains with the owner of the concession until the company has bought its compulsory train (cf. 4.4.7) or becomes insolvent (cf. Rule 6) – depending on which event occurs first.

3.2.2 SALE OF SHARES (NOT ALLOWED IN SR1)

In this action, a player sells one or more shares from their assets to the bank. All shares they want to sell from one company must be sold as a lot before shares in another company may be sold.

The sold shares are placed in the bank pool, to distinguish them from new shares that were not yet bought by players.

For each share sold, the player receives its current stock value from the bank. This is the stock value on the stock market *before* the sale.

Exception: If a company has not yet operated, selling a share in this company gives the selling player only the stock value that is one space lower on the stock market than the share's current stock value.

Selling of shares reduces the company's stock value on the stock market by one space per share sold.

There are a number of exceptions to this rule:

- If a company has not yet operated, its stock value is not affected by sales.

- The stock value can never drop below 50.
- The stock value of shares in the Magdeburg-Halberstädter Eisenbahn (MHE) is not affected by sales once the MHE has bought a 5-train (cf. 3.2.7 for details).

Regardless of whether the stock value of a company whose shares were sold drops or remains the same, the company's stock value marker is always placed underneath any stock value markers of other companies whose markers may already occupy the same space of the stock market.

Sales Restrictions:

- Shares may not be sold in the first stock round.
- The director of a railway company that has not yet operated cannot sell below 20% of this company.

Otherwise, the director of a public company can sell as many shares as desired provided at least one other player possesses at least 20% of this company. If, after such a sale, at least one other player has more shares in the company than the previous director, a change of directorship occurs. The player with the most company shares becomes the new director. If several players are tied for the percentage of ownership, this is resolved in current turn order starting from the previous director. (However, in the case described as an exception in 3.2.1, there is no immediate change of directorship!)

Clarification: Up to 80% of shares in a company may be in the bank pool. Companies in receivership always have 100% of their shares in the bank pool.

3.2.3 PURCHASE OF A SHARE

A player may buy *one* share of a public company at the current stock value from the bank pool or initial offering if the following are true:

- The company is not in receivership
- A share is on offer
- The player has not sold shares in this company in this SR.

The stock value of the company is *not* affected by this action.

Shares that are on offer are either in the initial offering or in the bank pool. The available MHE shares are always in the bank pool. If a company has shares both in the initial offering and the bank pool, the player decides which one they want to purchase.

When **buying a share from the bank pool**, the player pays the current stock price to the bank and receives the share.

When **buying a share from the initial offering** of a company, there are two cases:

- If the company has not yet operated, the player pays the current stock price to the bank and receives the share. The paid money is held by the bank until the company floats; then it is fully capitalized by the bank (cf. 3.2.3.1).
- If the company has already operated in an OR, shares in the initial offering are a result of the issue of new shares (cf. 4.3.6 and 4.4.8). The player pays the stock price to the company treasury and receives the share.

If, after the purchase of a share, the buying player has more shares in a company than the company's director, there is now a **change of directorship**, except in the special case described in 3.2.1.

3.2.3.1 FLOTATION OF A COMPANY

A railway company floats as soon as the third of its original shares is bought (cf. 3.2.3) from the initial offering. The following steps are successively completed prior to the next player action in the stock round:

- The owner of the concession receives the company charter, on which all the company's assets are kept: its treasury, its trains and switchers, and its unused station markers.
- The company receives as starting capital five times its current stock value from the bank. This is the equivalent of 100% of its initially issued shares.
- The two shares of the company that are still in the initial offering are now placed in the bank pool.

Then the owner of the concession constructs the company's home stations by placing station markers in all station circles containing the company's herald at no cost.

Clarification: Most of the railway companies have two home stations, which are at the endpoints of their respective concession routes. Exceptions are the NWE, with a third home station on its concession route, and the QLB, which has no concession route and only a single home station (cf. Table 2).

In the next OR, the company operates for the first time.

3.2.4 FORMATION OF A RAILWAY COMPANY

A player can take this action only if they own the concession of the respective railway company and has enough money to purchase at least one of the company's shares. The formation procedure consists of a number of steps carried out in the following order:

- The player selects the **initial stock value** for the company. The allowed initial stock values are shaded on the stock market: 120, 150, 190, 240, and 300 Marks.
- Mark the initial stock value on the stock market with the company's stock value marker (one of the company's markers with a blank side). The marker goes below any existing markers at that price.
- Place the five 20% shares of the company in its initial offering space.
- The player must now immediately buy one of these shares. If they have enough money, they may purchase a second share in the same action.

Clarification: A railway company does not automatically float when formed. Flotation happens only when the third share is purchased from the initial offering (cf. 3.2.3.1.). A railway company can float in the same stock round it is formed, or in a subsequent stock round, depending on when the third share is bought.

3.2.5 FORMATION OF A PUBLIC MINING COMPANY

This action is not available until the second stock round (Exception: Harzer Werke: cf. 3.2.5.1). A player may take this action only if they possess at least one independent mine, there is at least one other independent mine available for the formation, and there is at least one public mining company that has not yet formed. The formation is done through a merger of two independent mines. One of them must be owned by the current player; the second one may belong to another player, provided they agree on this action. Closed mines may *not* be used for the formation of a public mining company.

Clarification: Two arbitrary open independent mines may be used to form one of the following four public mining companies:

- Concordia (CO)
- Montania (MO)
- Schachtbau Nordhausen (SB)
- Union (U)

Also Vor-Harzer mines (cf. Table 1) are allowed. For the Harzer Werke (HW) there are special rules (cf. 3.2.5.1).

The formation steps are carried out in the following order:

- The player chooses a yet unformed public mining company and takes its charter.
- They identify the two independent mines that merge to form the public mining company. These mines are the **initial mines** of the company.
- The **initial stock value** of the company is marked on the stock market with the company's stock value marker. It is placed below any existing markers in that space. The initial stock value is the average of the face values of the initial mines rounded down to the next valid stock value.
- The player places the certificates of the two initial mines on the company charter of the new public mining company in the designated spaces. These two mines are now the property of the public mining company. Money and switchers owned by the initial mines are also transferred to the new mining company and are placed on its charter. The machine markers of the initial mines remain with the respective mines and cannot be moved between the mines of the company.
- For each of the two initial mines, their former owners get in exchange a 50% share of the public mining company. The current player is its first director, even if the second initial mine comes from a different player.

The company operates for the first time in the next OR.

Example 1: Mines #15 (face value 300) and #7 (face value 170) are merged into a public mining company. Half of the sum of the face values is 235. The initial stock value is 220.

Example 2: Mines #13 (face value 260) and #2 (face value 120) are merged into a public mining company. The initial stock value is 190.

3.2.5.1 SPECIAL RULES FOR THE FORMATION OF THE HW

All rules for the formation of public mining companies apply also to the Harzer Werke AG (HW), with the following exceptions:

- The HW is the only public mining company that can be formed in the first stock round, but only by the owner of Mine #12 (Blankenburg), and only if the Blankenburg mine is chosen as one of the initial mines of the HW. If the owner of the Blankenburg mine does not form the HW during the first stock round, this privilege to an early start is forfeited.
- Only mines designated as "Vor-Harzer" mines (cf. Table 1) are allowed to form the HW.

Clarification: If the owner of the Blankenburg mine does not form the HW during the first stock round, then from the second stock round on, the HW can be formed by any player owning a "Vor-Harzer" mine. The Blankenburg mine need no longer be one of the initial mines, though both initial mines must still be "Vor-Harzer" ones.

3.2.6 EXERCISE OF A PURCHASE OPTION

This action can be performed by the current player in a stock round if they have acquired a purchase option for a company in receivership in the preceding auction round. To exercise this option, they buy at least 20% and up to 40% of the shares of the company in a single stock purchase action. If the company has a 50% stock split, the purchase option is exercised by buying a 50%-share.

Once this action has been performed, the company is released from receivership with the acting player as its new director.

The remaining shares of the company are now available for purchase in normal purchase actions (cf. 3.2.3).

3.2.7 SHARES IN THE MHE

The operation of the Magdeburg-Halberstädter Eisenbahn (MHE) happens largely outside the area of the game. Nevertheless, the players can buy and sell shares of the MHE, but no player is ever the director as the MHE shares available in the game represent only a small portion of all the MHE shares.

The game has 10 MHE shares. They can be purchased beginning in the first stock round (cf. 3.2.3). The initial stock value is 120.

From the second stock round they may also be sold (cf. 3.2.2), with the stock value dropping according to the normal rules, until the MHE buys a 5-train. Thereafter, the stock value of the MHE remains fixed for the rest of the game and is no longer affected by selling.

3.2.8 END OF A STOCK ROUND/DETERMINATION OF TURN ORDER

Once all players have successively passed, the stock round ends.

Any concessions in players' hands that were not used in the present stock round to form the respective railways are now returned to the bank without compensation. They are again available for auction in the next auction round.

Any purchase options not exercised during the present stock round are also forfeited without compensation and are again available in the next auction round.

Turn Order for the next trading round is now determined. The player with the most cash receives the turn order card #1, the player with the second-most cash gets the turn order card #2 and so on. If two players have the same amount of cash, the prior turn order between these two players is retained.

The newly determined turn order applies throughout the next trading round. In particular, this means that the player with turn order card #1 is the first one to act in both the next auction round and the next stock round.

At the end of the stock round, the trading round is complete and a set of one to three operating rounds follows. The number depends on the game phase (cf. Rule 5).

4. OPERATING ROUNDS (ORs)

In each OR, first the unmerged independent mines operate in ascending order of their face values (cf. 4.2).

Then the public companies operate in descending stock value order (cf. 4.3 and 4.4). If two or more companies have the same stock value, the company with the topmost stock value marker goes first, the others following in descending marker order.

The MHE is not subject to this rule and always operates last in every OR (cf. 4.5).

4.1 MACHINES AND ROLLING STOCK (TRAINS, RAILCARS, SWITCHERS)

In order to generate revenue, independent and public mining companies need to own machines and railway companies need to have trains. There are also switchers (also known as shunters) which are small locomotives that move wagons around in the mines and to the railway lines. They further raise the income of mining companies. Towards the end of the game, the railway companies can also acquire diesel railcars.

Machines can only be owned by mining companies. Trains (including diesel railcars) can only be owned by railway companies. Switchers can be owned by companies of any kind.

Machines, trains and switchers come in several "sizes"

- Machines (M1-M5)
- Switchers (S2-S5)
- Trains (1-5)

Starting with size 1, they become available in succession over the course of the game. For any size, there are a specific number of **units**. Beginning with size 2, units can be purchased as either a machine or a train. (For switchers of any size and for machines and trains of size 1 special rules apply – see below.) Units of the next size are available only once all units of a particular size have been purchased.

Units of size 1: All mines begin with a machine of size 1, and the MHE starts with a size 1 train. There is only one other unit of size 1. This unit can only be purchased as a train.

Units of size 5: There are an unlimited number available.

Diesel railcars: After the beginning of the first full set of three ORs in Phase 5 (cf. Rule 5), diesel railcars become available. These are special units of size "D" that can only be purchased by railway companies. (Machines, trains and switchers of size 5 remain available until the end of the game.)

Standard trains: Trains of sizes 1 to 5 that are not diesel railcars or switchers.

Switchers: Switchers are available in sizes 2 to 5 (in any quantity) and are not limited by the number of switcher markers contained in the game. Switchers do not count against the number of available units. Switchers of size n become available once the first unit (standard train or machine) of size n has been purchased. They remain available until the first machine or train of the next size has been bought. (Consequently, starting with Phase 2, there are always switchers of exactly one size available.)

TABLE 3. MACHINES/TRAINS/SWITCHERS

UNIT SIZES	NUMBER AVAILABLE*	PRICE AS MACHINE	PRICE AS TRAIN	PRICE OF SWITCHER**
1	1	--	100	--
2	10	150	250	50
3	7	300	450	100
4	3	500	750	150
5	Unlimited	800	1200	200
D	7	--	250***	--

* Only applicable to trains and machines.

** Purchase does not reduce the number of available units.

*** Diesel railcar.

The following restrictions must be followed:

- All standard trains and diesel railcars are only tradable between railway companies.
- Each railway company may own up to one diesel railcar at a time.
- Each independent mine may own up to one switcher at any time.
- The maximum number of switchers a public mining company may own at any time is equal to the number of mines it owns.
- Railway companies may own any number of switchers, but they cannot use them to generate revenue.
- Switchers are tradable between independent mines, public mining companies, and railway companies.
- Machines are not tradable between companies.

4.2 OPERATION OF INDEPENDENT MINES

The turn of an independent mine consists of several steps that are performed by its owner in the following order:

- Track Building (cf. 4.2.1)
- Production (cf. 4.2.2)
- Profit Distribution (cf. 4.2.3)
- Purchase and Scrapping of Machines and Switchers (cf. 4.2.4)
- Closing a Mine (cf. 4.2.5)

Clarification: All independent mines start with a treasury of 0 Marks.

4.2.1 TRACK BUILDING

Each OR, an independent mine that is not yet connected to the state railway network can build a single track tile or (from Phase 3 on) perform an upgrade. The usual rules for track building apply (including the rules for construction costs, cf. 4.4.1.) except routes for independent mines may be traced from any location within their hex instead of a station marker.

A mine is considered **connected** to the state railway network if there is continuous track from the mine's hex to some state railway track. Mine #15 (Thale) is therefore already connected at the beginning of the game.

Reimbursement of construction costs: If an independent mine has to pay construction costs when building track on a hex containing a yet unbuilt part of the concession route of a railway company, then, immediately after the independent mine has paid the construction costs, its owner is reimbursed an equal amount of money from the bank. The reimbursement is paid to the mine's player owner. The concession of the corresponding railway company may thus be activated (cf. 4.6).

Building restrictions: Track building on hexes that contain a yet unbuilt part of a concession route of a railway company is only allowed if either exactly this part of the concession route is built or, else, if the build doesn't prevent the complete concession route from being built. In the latter case, an appropriate track tile must be reserved for this hex that maintains the existing track and adds just the track yet to be built to complete this part of the concession route. If all tiles that meet this condition are presently built or if there is no such tile at all, the desired build cannot be performed.

4.2.2 PRODUCTION

In this step, the **revenue** produced by the mine is determined. The amount depends on several factors:

- Whether the mine is connected to the state railway network
- What size its machine has
- Whether it owns a switcher
- The size of such a switcher

If the mine is not yet connected to the state railway network (cf. 4.2.1), the mine's certificate has its "unconnected" face up, i.e., the face with the track symbol crossed out. The number below the machine marker space is then the revenue produced by the

mine, regardless of the size of its machine and the possession of a switcher.

As soon as the mine is connected to the state railway network, its certificate is flipped over to its "connected" face, indicated by the non-crossed track symbol. If the mine has a machine marker of size greater than 1, it is placed in the accordingly numbered space. If it owns a switcher, its switcher marker is similarly placed. The mine's revenue is then the number below the machine marker (or below the space M1 if it has only a machine of size 1) plus, if applicable, the number above its switcher marker. So, in contrast to unconnected mines, a connected mine may have higher revenue if it owns a higher-sized machine and/or a switcher.

The revenue is then reduced by due **maintenance costs** starting in Phase 3. The amount of maintenance cost depends on the game phase and item (cf. Rule 5). If the revenue is lower than the maintenance costs, reduce the treasury of the mine accordingly. If the maintenance costs cannot be fully paid, the mine is closed without compensation. Such a **forced closure** is handled in the same way as a voluntary closure (cf. 4.2.5).

What remains of the revenue after reduction by the maintenance costs, is called the mine's **profit**. The profit is always considered to be zero if the revenue is less than or equal to the maintenance costs.

The Player's Guide contains detailed examples for the calculation of revenue and profit of mines.

4.2.3 PROFIT DISTRIBUTION

The bank then pays 50% of the profit to the owner and the other 50% into the treasury of the independent mine. Place the money nearby or overlapping the independent mine card so that it stays separate from other money piles.

4.2.4 PURCHASE AND SCRAPPING OF MACHINES AND SWITCHERS

If the company has enough money in its treasury, it can now purchase a new machine (thus scrapping its existing one) and/or switchers (also scrapping its existing one). Purchases and scrapping can be performed in any order.

- Machines can only be purchased from the bank. Transfer of machines between companies is not allowed.
- A machine can only be purchased if it is of a higher size than the mine's current machine.
- When purchasing a machine, the mine gets *one* machine marker of the new size. Its existing machine is automatically scrapped (at no cost) and removed from the game. If the mine is not yet connected to the state railway network, the new marker is placed in the M1-space of the mine's card. Otherwise, it is placed onto the M-space corresponding to its size.
- Machine markers can never be voluntarily removed. A mine retains its machine until it is replaced by a new machine or for the rest of the game (provided the mine is not closed).
- Subject to the limitation that the new machine must be of higher size than the existing one, multiple machines may be bought successively.

Example: The mine has a 1-machine. It can buy the last 2-machine and then also the first 3-machine if it has enough money for these purchases.

- A switcher can (from Phase 2 on) be purchased from the bank for face value or from another company. When buying from another company, the price must be at least 1 Mark and not more than double face value. Maintenance costs may be due depending on the game phase (cf. Rule 5). If so, either the *buying* or the *selling* company must pay these maintenance costs to the bank as part of the purchase. The newly acquired switcher marker is placed either in the S-space of the mine's certificate, if the mine is not yet connected to the state railway network, or in the S_n -space if the switcher is of size n .
- More than one switcher may be purchased in a turn; however, with each purchase, any previously-owned switcher is automatically scrapped since an independent mine may only own *one* switcher at any time. Scrapped switchers are out of the game. Place the scrapped switcher's marker back in the pool of switcher markers.
- A mine may also scrap a switcher (at no cost) without simultaneously purchasing a new one.

4.2.5 CLOSING A MINE

As the last action of an independent mine, the owner may close the mine. They return the mine certificate to the bank and receive the mine's treasury (which is nothing in case of a *forced* closure). After the closure, the mine has no money in its treasury and only retains its inherent 1-machine. An existing switcher or higher-sized machine is removed. Its marker on the board is flipped to the "closed" side.

Later in the game, the closed mine may be purchased by a public mining company at face value from the bank. It can no longer be acquired by players.

4.3 OPERATION OF PUBLIC MINING COMPANIES

Clarification: Public mining companies are not allowed to build or upgrade railway track.

The turn of a public mining company consists of a number of steps that are performed by its director in the following order:

- Production (cf. 4.3.1)
- Profit Distribution (cf. 4.3.2)
- Stock Value Adjustment (cf. 4.3.3)
- Purchase a Mine (cf. 4.3.4)
- Purchase and Scrapping of Machines and Switchers (cf. 4.3.5)
- Issue New Shares (cf. 4.3.6)

4.3.1 PRODUCTION

The **revenue** produced by the public mining company is the sum of the revenues of its mines. If the company possesses switchers, they can be freely transferred between the mines for production. For mines that are not yet connected to the state railway network,

the switcher markers are placed in the S-spaces of the mine certificates on their "unconnected" faces. Otherwise, n -switchers are placed in the S_n switcher spaces on the "connected" faces of the certificates, where n is the size of the respective switchers. Each switcher *must* be assigned to exactly one of the company's mines. No more than one switcher per mine is allowed.

The revenue is then reduced by the **maintenance costs**, depending on the game phase (see tables on the company charter or cf. Rule 5). If the revenue is lower than the maintenance costs, reduce the treasury of the mining company accordingly. If the maintenance costs cannot be fully paid, the mining company becomes insolvent (cf. Rule 6).

What remains of the revenue after reduction by the maintenance costs, is called the company's **profit**. The profit is always considered to be zero if the revenue is less than or equal to the maintenance costs.

4.3.2 PROFIT DISTRIBUTION

The director determines whether the profit is withheld, half distributed or completely distributed. When withheld, the bank pays the entire profit into the company's treasury. In the case of half distribution, the bank pays half of the profit to the company's treasury and the other half is distributed to the shareholders. Dividends are paid to the shareholders according to their respective shares in the company. If the distribution amount for a player is not a whole number, the bank rounds in favour of the player to the next whole number.

Clarification: If a dividend is paid, the dividend for shares that are not held by players remains in the bank, regardless of whether these shares are in the initial offering or in the bank pool.

4.3.3 STOCK VALUE ADJUSTMENT

One of the following stock price adjustments is made:

- If the profit is withheld or zero, then the stock value of the company drops one space on the stock market.
- If the distributed amount is greater than zero but lower than the company's current stock value, then the stock value remains the same.
- If the distributed amount is at least equal to the current stock value but lower than twice the current stock value, then the stock value rises one space on the stock market.
- If the distributed amount is at least twice the current stock value but lower than three times the current stock value, then the stock value rises two spaces.
- If the distributed amount is at least three times the current stock value, then the stock value rises three spaces.

The stock value cannot rise more than three spaces on the stock market through a profit distribution.

Regardless of whether the stock value of a company drops, rises or remains the same in this step, the company's stock value marker is always placed underneath any stock value markers of other

companies whose markers may already occupy the same space of the stock market.

4.3.4 PURCHASE A MINE

Under certain conditions, the public mining company may now buy either an open mine (from a player) or a closed mine (from the bank). Mines may never be bought from other public companies. Every public mining company can buy at most one mine per OR.

When the mine is bought from a player, the purchase price must be at least 1 Mark and not more than double face value. When it is bought from the bank, the purchase price is always the face value.

The purchased mine together with all its assets becomes property of the buying company, i.e., the mine retains its machine marker, and its treasury and switcher (if the mine has one) are transferred to the company's possession.

A closed mine is automatically reopened when purchased, but only with its inherent 1-machine. (Closed mines have never a higher-sized machine, nor do they have any treasury or switcher.) If a mine is reopened in this way, its mine marker on the map is flipped over to its active side.

The following conditions must be fulfilled for the purchase of a mine:

- An independent mine must be available, i.e., one that is not yet part of a public mining company.
- The company must have enough money to pay the purchase price.
- The company must have a free slot for the mine on its company charter: Companies with a 50% stock split may possess only two mines, those with a 20% stock split not more than four mines, and those with 10% stock split up to five mines.
- Buying a mine from another player is only possible if the owner of the mine consents and both players agree upon the purchase price.

4.3.5 PURCHASE AND SCRAPPING OF MACHINES AND SWITCHERS

If the company has enough money in its treasury, it can now purchase one or more new machines (thus scrapping existing ones) and/or switchers; also, switchers may be scrapped. Purchases and scrapping can be performed in any order. The following rules apply:

- Machines can only be purchased from the bank. Transfer of machines between companies is not allowed.
- A new machine must be of higher size than at least one current machine marker of the company.
- When purchasing a machine of size n (where n is 2, 3, 4, or 5), the company gets up to n machine markers of size n . Each of the new machine markers must replace an existing machine marker of lower size. The replaced machine markers are scrapped (at no cost) and removed from the game. If a mine is not yet connected to the state railway network, the new marker is placed in the M1-space of the mine's certificate. Otherwise it goes into the M-space corresponding to its size.

- Machine markers can never be voluntarily removed. Every mine retains its machine marker until it is replaced by a new machine marker or for the rest of the game.
- Switchers can (from Phase 2 on) be purchased from the bank for face value or from another company. When buying from another company, the price must be at least 1 Mark and not more than double face value. When a switcher is bought from another company, maintenance costs may be due (refer to the table on the company charter and cf. Rule 5). If so, either the *buying* or the *selling* company must pay these maintenance costs to the bank as part of the purchase. When buying a switcher, the company gets a switcher marker of the corresponding size.
- At any time, a public mining company may not own more switchers than mines. So, surplus switchers are automatically scrapped. Scrapped switchers are out of the game. Place the scrapped switcher's marker back in the pool of switcher markers.
- Switchers may also be scrapped (at no cost) without purchasing new ones.

4.3.6 ISSUE NEW SHARES

If the company has a stock split of 50% or 20%, it can now issue new shares. A pre-condition is that all previously issued shares must be held by players. There are two cases:

- A company with a 50% stock split issues three new shares. All shares of the company are now 20% shares.
- A company with a 20% stock split issues five new shares. All shares of the company are now 10% shares. In order to execute this action, the director must own at least two 20% shares of the company, i.e., a minimum of 40%.

The newly issued shares are placed in the initial offering. When they are later purchased for the first time, the current stock price has to be paid, giving the money to the company's treasury.

4.3.7 SPECIAL RULES FOR THE HW

The Harzer Werke (HW) operates in the same way as the other public mining companies, with the following exceptions:

- During the production step, the HW always gains an additional revenue of 50 Marks from smaller mines that are not explicitly represented in the game. The respective "HW +50" marker should be placed on the HW charter as a reminder of this revenue.
- When purchasing a mine, the HW is restricted to the Vor-Harzer mines (cf. Table 1 and the mine certificates).

4.4 OPERATION OF RAILWAY COMPANIES

The operation of a railway company consists of a number of steps that are performed by its director in the following order:

- Track Building (cf. 4.4.1)
- Station Construction (cf. 4.4.2)
- Running the Trains (cf. 4.4.3)
- Calculation of Revenue (cf. 4.4.4)
- Profit Distribution (cf. 4.4.5)

- Stock Value Adjustment (cf. 4.4.6)
- Purchase and Scrapping of Trains and Switchers (cf. 4.4.7)
- Issue New Shares (cf. 4.4.8)

4.4.1 TRACK BUILDING

Track is built by placing the hexagonal **track tiles** on the hexagonal grid of the map. The individual spaces of this grid are called **hexes**. Track has already been constructed in some of the hexes. Thick black lines represent track of railway companies that operate more or less outside the game. For game purposes such track is called **state railway track**. Such track may not be used by any of the railway companies of this game. Black and white dashed lines represent existing narrow gauge track that *may* be used later by the railway companies. Thin black lines do *not* represent track, but they depict the paths of the **concession routes** that need to be built by the respective railway companies. In most hexes with preprinted track, there are also circles called **station circles**, in which station markers can be placed (cf. 4.4.2). The track tiles show track that can be built and many of the tiles have also station circles.

Track tiles come in four different colours: yellow, green, brown, and grey. Which ones of these tiles are currently available for track building depends on the Game Phase (cf. Rule 5):

- At the beginning of the game, only *yellow* track tiles may be used for track building. Yellow tiles may only be placed on the light green or light brown hexes of the map that do not already contain track, and the first track tiles to be placed on these hexes must always be yellow ones throughout the game. On the three light brown concession-route hexes of the WBE (C10, C12, C14), track may only be built from Game Phase 3 on. Yellow tiles are considered to be unlimited in quantity.
- Once Game Phase 3 has started, the *green* tiles become additionally available. These may only be used to replace already built yellow tiles or to be placed on yellow preprinted hexes.
- Beginning in Game Phase 5, the *brown* tiles become available. These may only be used to replace already built green tiles or to be placed on green preprinted hexes.
- Finally, as soon as the first diesel railcar is bought, the *grey* track tiles come into play. They may only be used to replace previously built brown tiles.

Placing green, brown, and grey track tiles on the map is also called **upgrading**. Replaced tiles can again be used for building.

For placing and upgrading track tiles, the following rules apply:

- Track tiles *without a town or village* (i.e., without a station circle) may only be placed on hexes without a town or village.
- Track tiles *with a village* may only be placed on hexes with a village. On the light green and light brown map spaces, villages are shown as thick black dots. On yellow, green, brown, and grey map spaces and track tiles, villages are represented by one to three station circles. There is always only *one* village per hex, regardless of the number of station circles, even if the station circles are not connected. Village hexes and tiles are those ones with station circles that neither have a purple border nor contain state railway track.

- Track tiles *with a town* may only be placed on hexes with a town. Town hexes as well as the track tiles that can be placed on them have a purple border.
- The special town tiles marked with a “B” may only be placed on the Blankenburg hex (D15).
- The special town tiles marked with “HQG” may only be used for the hexes Halberstadt (B19), Quedlinburg (E20), and Gernrode (G20).
- Tiles must *not* be placed so that newly built track runs against an outer edge of the map’s hex grid, against one of the thick black barriers or against an edge of a grey map space when there is no track connection.
- Tiles may never be placed on grey map spaces.
- When upgrading, existing track must be maintained and existing connections must not be cut off.
- Mine markers and station markers remain in their respective positions as upgrades are performed.
- Placing or upgrading a track tile is only permitted if *all new* track is accessible for the building company, even when upgrading towns or villages. It is *not* sufficient that the value of the station rises. Here, “accessible” means that, *after* the placement of the new tile, starting at an active or inactive station marker of the company, there must be a continuous track connection to *all* new track that neither contains any state railway track nor a town or village that is completely blocked by station markers of other companies so that the connection is open to the company. This even applies to the hex where the upgrade takes place if the new track is *behind* the upgraded town or village. (Cf. the examples in the Player’s Guide.)
- Unconnected station circles of a village become connected when upgraded from green to brown. This also applies to the green map spaces Sorge (F5) and Tanne (F7), on which, from Game Phase 5, arbitrary brown village tiles may be placed that maintain the existing track. I.e., green hexes F5 and F7 and green tiles #965–970 are upgraded with appropriate brown tiles #975–977.

Construction costs: In plain terrain, tiles are placed at no cost. In more difficult terrain, construction costs must be paid when placing a tile on a respective map space. The construction costs are shown on the map and represent hilly, mountainous and high mountainous terrain. The construction costs are due when placing the first tile on a hex with a mountain symbol, regardless of the colour of the map space. Once a tile has been placed on a hex, further upgrades incur no further construction costs. The company building the tile pays the relevant construction costs to the bank.

Reimbursement of construction costs: If a railway company has to pay construction costs when building track on a hex containing a yet unbuilt part of the concession route of *another* railway company, then, immediately after the building railway company has paid the construction costs, its director is reimbursed an equal amount of money from the bank. The reimbursement is paid to the

director. (The concession of the corresponding railway company may thus be activated, cf. 4.6 and Table 2.)

Building restrictions: Track building on hexes that contain a yet unbuilt part of a concession route of *another* railway company is only allowed if either exactly this part of the concession route is built or if the building doesn't prevent the complete concession route from being built. In the latter case, an appropriate track tile must be reserved for this hex that maintains the existing track and adds just the track yet to be built to complete this part of the concession route. If all tiles that meet this condition are presently built or if there is no such tile at all, the desired build cannot be performed.

If the corresponding company later builds its concession route, such reserved tiles are then used even if tiles of that colour are generally not yet available in the current game phase.

Construction of the concession route: In its first OR, a company must build its complete concession route (unless it was completely built before). All track tiles necessary to complete the concession route must be placed in one single building action. The complete construction costs for the concession route (cf. Table 2) must be paid, even if for some or all of the relevant hexes construction costs were previously paid by other companies. (In such cases, the bank has funded the construction costs in advance and now demands its money back.)

Track building after completion of the concession route: In its first OR, a company is not allowed to build track in addition to completing its concession route. In later ORs (and also in the first one if the concession route was completely built in previous ORs), it may place or upgrade one or two tiles. Two tiles are only allowed if *both* are from the following set: yellow tiles or green tiles with two unconnected track sections, i.e., tiles #75–79, #914, #956–961, and #964–970.

4.4.2 STATION CONSTRUCTION

Stations of railway companies are represented by their station markers placed on the map. Each railway company has a specific number of station markers available, of which only a part may be used if the company's shares have a 20% denomination (cf. Table 2). The station markers have two faces, an *active* one and an *inactive* one. Normally, the station markers have their active face up.

When a railway company floats (during a stock round), it immediately places its own station markers in each of its home station circles at no cost (cf. 3.2.3.1). (No other companies are ever allowed to build stations there.) Additional stations can only be built during operating rounds, where each railway company can, during its turn, build at most one station per OR. All station markers that are newly placed on the map are placed with their active face up.

The construction of a new station is only possible if the company still has a station marker available and has enough money to pay the construction costs.

In order to construct a station for a railway company, its director takes one of the company's available station markers from the company charter and either places it in an empty station circle

on the map or replaces an inactive station marker of another company with it. The placement in an empty station circle costs 100 Marks; the money is paid to the bank. The replacement of an inactive marker of another company costs 50 Marks. In this case, the money is paid to the other company. This company gets its replaced station marker back and may use it again for station construction during its own turn. For replacing another company's inactive station marker with its own marker, an agreement with the other company's director is *not* required.

The following rules apply for the construction of stations:

- The new station must be *accessible* to the railway company that builds the station; i.e., there must be a continuous track connection from a previously built (active or inactive) station marker of the company to the space where the new station is to be built, and this track connection must neither contain any state railway track nor a town or village that is completely blocked with station markers of other companies.
- Each railway company may have at most one station per hex.
- If a company has not yet operated, no other company is allowed to build a station in any village belonging to the concession route of that company, unless such a station building does not prevent the company from building a station in the same hex of its concession route during its first OR.
- Station markers may never be voluntarily removed from the map. Only by replacement of an inactive station marker with another company's marker may a company get a station marker back.

4.4.3 RUN TRAINS

If the company has at least one standard train, the director now determines the routes on which all the company's trains run. (For companies in receivership special rules apply, cf. 6.2.) Switchers *cannot* be run by railway companies. State railway track may *never* be part of a route.

A **route** is a linear stretch of track that begins at a station marker (active or inactive) of the operating company, a factory, or an open mine, and ends at some other station marker (active or inactive) of the company, some other factory, or another open mine. The route must contain at least one (active or inactive) station marker of the company. A route may not pass through a town or village that is completely blocked by station markers of other companies (except for the company's own concession route) nor may it end in such a town or village, even if there is a factory or an open mine in the hex. Routes are, in principle, of any length, but must end once they enter a town hex (i.e., one with a purple border). No route may contain a hex with a town or village more than once.

*Clarification: Multiple station circles in one hex, even when unconnected, always represent only **one** town or village.*

A train of size n (where n is 1, 2, 3, 4, or 5) can run up to n routes, but these routes must be directly or indirectly connected, thus forming a connected network. Two routes are considered directly connected if they meet in some town or village that is contained

in both routes; they are considered indirectly connected if they are connected via a sequence of mutually directly connected routes belonging to the network.

Clarifications:

(1) Two routes that “meet” in a village with two **unconnected** station circles are **not** considered to be directly connected.

(2) It is **not** necessary to have a company’s own station marker in the towns or villages where routes meet; nevertheless, such towns or villages must not be completely blocked by station markers of other companies.

(3) The routes actually serviced by the train must form a connected network in the above sense. It is **not** sufficient to run a selection of non-connected routes from a connected network. The serviced networks of **different** trains need not be connected with each other, though.

The concession route of a railway company must always be serviced by one of the company’s standard trains. Service by a diesel railcar is not sufficient. The concession route of the NWE consists of two connected routes in the above sense. These two routes must always be fully serviced, either by a single train of size at least 2 or by two trains, each one servicing one of the two routes.

No two of the routes serviced by standard trains of a company may have any track in common. They may, however, cross each other or meet at towns or villages. No route may be serviced by more than one standard train of the company.

If the company owns a diesel railcar, it operates in addition to and independent of the operation of the standard trains of the company in the way specified below.

Since the routes a diesel railcar services represent passenger transport only, the diesel railcar routes of a company must always start and end at station markers owned by the company.

A diesel railcar of a company always services the company’s concession route, and moreover any number of further diesel railcar routes of the company that are directly or indirectly connected with the concession route. (For the QLB, this means that Quedlinburg must be the endpoint of at least one of the railcar’s routes.) Nevertheless, as with the routes a standard train may run, the routes a diesel railcar services may cross each other or meet in towns or villages but are not allowed to have any track in common.

The total network of all routes that are serviced according to the conditions above, is called the *serviced network* of the company.

When running trains, as many routes as possible must be serviced. All routes must be serviced as far as possible. Maximizing the revenue generated, however, is *not* required.

All station markers of the operating company contained in the network just serviced by the company are now considered to be active. They are turned over to their active face if they were inactive before. All other station markers of the company become

or remain inactive and are turned over to their inactive face if they were active before. However, a company’s station markers on its own concession route, especially its home station markers, never become inactive. Inactive station markers may still block routes and track connections.

4.4.4 CALCULATION OF REVENUE

By servicing routes as described in 4.4.3, a railway company achieves **revenue**, which is the sum of the following values:

- The values of all **stations** of the company (i.e., villages and towns provided with its own station markers) that are run to by a standard train. The values of stations in grey map hexes with a multi-coloured value-strip depend on the current game Phase (yellow: Phases 1–2; green: Phases 3–4; brown: Phase 5; grey: Phase D). Each of these stations is only counted once, even if contained in more than one route or run to by more than one standard train.
- For each **stop** (i.e., town or village without its own station marker) run to by a standard train: 10. Each of these stops is only counted once.
- For each village completely blocked with station markers of other companies: 0 (also for diesel railcars). This case can only occur if a village on the company’s concession route is blocked.
- For every **station** or **stop** run to by a **diesel railcar**: 10. This is in addition to revenue obtained by standard trains.
- For every open **mine** run to by a standard train: the product of the mine’s multiplier (i.e., the number in the lower section on the mine marker: 10, 20 or 30) and the size of the largest standard train running to the mine. Every such mine is only counted *once*. Mines in hexes blocked by other companies’ station markers are *not* counted. There is no revenue for *closed* mines.
- For each of the following **factories** run to by a standard train: sugar refinery Derenburg (B13), paper mill Weddersleben (E18), quarry Knaupsholz (C6), sawmill Hasselfelde (F11), quarry Unterberg in the Eisfelder Talmühle hex (H9): the specified value (40, 50, 60 or 70). The size of the trains running to such factories does not matter. Every serviced factory is only counted *once*. Factories in hexes blocked by other companies’ station markers are *not* counted.

Clarification: At the location of the quarry Knaupsholz there is no village, so a train running there generates only the 60 M revenue for the quarry itself.

Examples for the running of trains and the calculation of revenue can be found in the Player’s Guide.

After having calculated the revenue, subtract the **maintenance costs** for the company’s standard trains and switchers, depending on the game phase (see tables on the company concessions or cf. Rule 5). If the maintenance costs exceed the revenue, the difference must be made up from the company treasury. If the maintenance costs cannot be fully paid, the company becomes insolvent (cf. Rule 6).

What remains of the revenue after reduction by the maintenance costs, is called the company's **profit**. The profit is always considered to be zero if the revenue is less than or equal to the maintenance costs.

4.4.5 PROFIT DISTRIBUTION

This works just as in the case of public mining companies (cf. 4.3.2).

4.4.6 STOCK VALUE ADJUSTMENT

This too works just as in the case of public mining companies (cf. 4.3.3).

4.4.7 PURCHASE AND SCRAPPING OF TRAINS AND SWITCHERS

In this step, the company may buy trains and switchers; toward the end of the game diesel railcars are also available. In addition, the company's trains and switchers may be scrapped at no cost.

In its first OR, a railway company must purchase a standard train (with the exception of the QLB, cf. 4.4.9), the so-called **compulsory train**. The purchase of the compulsory train must be the first purchase of the company; switchers or a diesel railcar may only be purchased afterwards. The compulsory train can be purchased for its face value from the bank or at a mutually agreed upon price (at least 1M, at most double face value) from another railway company.

Special feature: To ensure that the NWE can service its complete concession route, it must not buy the 1-train as its compulsory train.

If a railway company does not have enough money to buy a compulsory train, it becomes insolvent and goes into receivership (cf. Rule 6). The director may refuse offers to buy a train from another company even if this would avoid insolvency.

Immediately after the purchase of the compulsory train, the privileges and obligations of the owner of the concession end, and the concession is removed from the game. If another player now has more shares in the company than the former owner of the concession, directorship changes.

Special feature: Since the QLB has a built-in compulsory train, removal of the concession and change of directorship (if applicable) take place at the start of step 4.4.7 in the QLB's first OR.

After the purchase of the compulsory train, the director is free to choose, within step 4.4.7, for the company to buy more trains and/or switchers and also to scrap trains and/or switchers according to the rules specified below. Purchases and scrapping can be performed in any order. Trains and switchers can be bought from the bank (at face value) or from other railway companies. Switchers may also be bought from mining companies.

The following rules apply:

- There are no upper limits for the number of standard trains and the number of switchers a railway company may own.
- Every railway company may own at most one diesel railcar at any time.
- When buying a standard train, a switcher, or a diesel railcar from another company, the price must be at least 1 Mark and at most double face value.

- When a train or a switcher is bought from another company, you need to check whether maintenance costs are due (see table on the company concession or cf. Rule 5). If so, either the *buying* or the *selling* company must pay these maintenance costs to the bank as part of the purchase.
- The last standard train of a railway company must *never* be scrapped or sold: after the purchase of the compulsory train, the company must possess at least one standard train (NWE: a standard train of size at least 2) at any time, in order to be able to service its concession route. (This train need *not* be the very train that was originally bought as the company's compulsory train.)
- Once scrapped, trains and switchers are out of the game. Place the scrapped switcher's marker back in the pool of switcher markers.
- Diesel railcars may not be scrapped.

4.4.8 ISSUE NEW SHARES

If the company has a stock split of 20%, it can now issue new shares. Pre-conditions are that all previously issued shares must be held by players and that the director owns at least 40% of them.

The company issues five new shares such that all shares of the company are now 10% shares.

The newly issued shares are placed in the initial offering. When they are later purchased for the first time, the current stock price has to be paid, giving the money to the company's treasury.

The company also receives its remaining, previously not available, station markers (cf. Table 2).

4.4.9 SPECIAL RULES FOR THE QLB

The Quedlinburger Lokalbahn (QLB) has no concession route and only one home station (Quedlinburg, E20).

The QLB comes with a built-in compulsory train that may not be sold. This "local" train is considered as a standard train for which maintenance costs never become due.

The local train services the Quedlinburg hex only, giving the value of Quedlinburg (60 or 90) as revenue.

This servicing of Quedlinburg by the local train corresponds to the obligatory servicing of the concession routes by their respective railway companies. This means that additional standard trains that the QLB may possess need not run to Quedlinburg. But in case at least one such additional train *does* run to Quedlinburg, the value of Quedlinburg is counted twice for the revenue total (as an exception to the normal rules).

A diesel railcar owned by the QLB must always run to Quedlinburg.

4.5 OPERATION OF THE MHE

The Magdeburg-Halberstädter Eisenbahn (MHE) operates last in every OR. It does not operate in the same way as other companies. It merely pays a dividend to the shareholders and then buys a train if certain conditions are met as explained below.

The amount of the dividend is determined by the size of the train the MHE owns: the total amount is 100 times the size of the train.

This means, the dividend is 10M per share if the MHE owns a 1-train, 20M per share if it owns a 2-train, etc.

The share value of the MHE is then adjusted according to the normal rules.

At the start of the game, the MHE possesses a 1-train. Later, in the *last* OR of every set of ORs (consisting of 1 to 3 ORs depending on the game phase – see Rule 5), it acquires a new standard train from the bank at no cost, replacing its old train, even if it is of the same size. This is regarded as a purchase of a standard train so that a new game phase may be triggered. The old train is scrapped and removed from the game.

Once the MHE has acquired a 5-train, it stops buying trains and the share value doesn't change anymore, neither through sales of shares nor through payment of dividends.

The MHE never buys switchers or diesel railcars.

Other companies may not buy trains from the MHE.

The MHE never pays maintenance costs for its trains.

Remark: The routine purchase of new trains by the MHE also simulates the technological progress in the world outside the Harz Mountains.

4.6 SYNOPSIS OF THE RULES FOR THE CONCESSION ROUTES

In this game, most of the railway companies have a so-called concession route. This is a specific line the company is obliged to build and to service. A number of rules describe the rights and duties connected with the concession routes, particularly ensuring that the mentioned obligations can always be met. These rules are summarized here.

Track building on hexes that contain a yet unbuilt part of a concession route of a railway company is allowed to be performed by independent mining companies and other railway companies if either exactly this part of the concession route is built or, else, if the building doesn't prevent the complete concession route from being built. In the latter case, an appropriate track tile must be reserved for this hex that maintains the existing track and adds just the track yet to be built to complete this part of the concession route. If all tiles that meet this condition are presently built or if there is no such tile at all, the desired build cannot be performed.

If the corresponding company later builds its concession route, such reserved tiles are then used even if tiles of that colour are otherwise not yet available due to the current game phase (e.g. green tiles in Phase 2).

If a mine or railway company has to pay construction costs when building track on a hex containing a yet unbuilt part of the concession route of *another* railway company, then, immediately after the building company has paid the construction costs, its owner/director is reimbursed an equal amount of money from the bank. The reimbursement is paid to the owner/director.

*Clarification: The construction costs are **not** reimbursed when a company builds its **own** concession route.*

The concession of the corresponding railway company may thus be activated:

- The concessions of the KEZ, the SHE, and the NWE are normally activated in Game Phase 3. However, they become active earlier, in Game Phase 2, as soon as a tile is laid in one of their respective concession route hexes (by another railway company or an independent mine).
- In the same way, the concession of the WBE, normally activated in Game Phase 4, becomes active in Game Phase 3 as soon as a tile is laid in one of its concession route hexes (except the Blankenburg hex).

A company with a concession route must completely build all relevant track in its first OR (unless the concession route was completely built before). All track tiles necessary to complete the concession route must be placed in one single building action. The full construction costs for the entire concession route, according to the figures on the concessions and in Table 2 (page 5), must be paid, even if for some or all of the relevant hexes construction costs were previously paid by other companies.

In its first OR, a company is not allowed to build track in addition to completing its concession route, unless the concession route was completely built before. In the latter case, it *may* build track according to the general rules even in its first OR.

If a company has not yet operated, no other company may build a station in any village belonging to the concession route of that company, unless such a station building does not block the concession route. This means that immediately after the track building step in a company's first OR, there must be a free station circle in every village on the concession route so that the company has the opportunity to build its own station in any one of them during its first OR. An open station circle that does not belong to the concession route is *not* sufficient.

Station markers of a company on its own concession route, including its home station markers, never become inactive.

The concession route of a railway company can always be serviced by this company, even through villages completely blocked by station markers of other companies. However, there is no revenue for such villages nor for mines or factories located there.

Moreover, the concession route of a railway company *must* always be serviced by one of the company's standard trains once it bought its first train. Service by a diesel railcar is not sufficient.

Special feature: The concession route of the NWE consists of two connected routes in the general sense of the rules. These two routes must always be fully serviced, either by a single train of size at least 2 or by two trains, each one servicing one of the two routes.

If the company owns a diesel railcar, it must service the concession route.

5. GAME PHASES

The game is divided into six game phases. The first phase begins with the Game Start. Phases 2 through 5 start as soon as the first standard train or machine of the corresponding size is bought. The start of the sixth phase (Game Phase D) is triggered by the purchase of the first diesel railcar. At the beginning of each new phase, certain changes to the game come into effect. These concern:

- The number of operating rounds between two trading rounds (1 to 3)
- The types of track tiles that are available (yellow / green / brown / grey)
- The values of stations in grey hexes with multi-coloured value bars, where the values correspond to the colours of the available track tiles (cf. 4.4.4)

- The amount of the due maintenance costs for machines, trains and switchers
- The activation of concessions
- The availability of switchers
- The availability of diesel railcars

The details of all these effects are summarized in the table below. All effects listed there commence immediately with the change of the game phase. The only exception is the number of ORs between two trading rounds. This number does not change until the next set of ORs commences.

GAME PHASES

TRIGGERING EVENT	GAME PHASE	ORS PER SET	TILES AVAILABLE	MAINTENANCE COSTS	REMARKS
Game start	1	1	Yellow	--	• HBE and GHE concessions are active. • Harzer Werke AG may be started from SR1, other public mining companies from SR2.
Purchase of the first standard train or machine of size 2	2	1	Yellow	--	• NWE, SHE, and KEZ concessions may be activated. • 2-switchers available.
Purchase of the first standard train or machine of size 3	3	2	Yellow and green	• 1-machines: 50M • 1-train: 50M	• NWE, SHE, and KEZ concessions become active if not activated earlier. • WBE concession may be activated. • 3-switchers available.
Purchase of the first standard train or machine of size 4	4	2	Yellow and green	• 1-machines: 100M • 2-machines: 50M per mine • 1- and 2-trains: 100M • 2-switchers: 20M	• WBE concession becomes active if not activated earlier. • QLB concession becomes active. • 4-switchers available.
Purchase of the first standard train or machine of size 5	5	3	Yellow, green, and brown	• 1-machines: 100M • 2-machines: 50M per mine • 1- and 2-trains: 100M • 2-switchers: 20M	• 5-switchers available until the end of the game.
Purchase of the first diesel railcar	D	3	Yellow, green, brown, and grey	• 1-machines: 150M • 2-machines: 100M per mine • 3-machines: 50M per mine • 1-, 2-, and 3-trains: 150M • 2-switchers: 50M • 3-switchers: 30M	• Diesel railcars are only available after the beginning of the first full set of ORs in Phase 5. • 5-trains and 5-machines remain available until the end of the game.

6. INSOLVENCY AND RECEIVERSHIP

A company becomes insolvent as soon as one of the following events occurs:

- a) A railway company that does not have enough funds to buy its compulsory train becomes insolvent. The director need not accept offers of other companies to buy a train from them, even if by such a purchase insolvency could be avoided.

Clarification: In this case, the stock value of the company drops one space before the Recapitalisation Procedure of 6.1 commences.

- b) A company has lower revenue than its maintenance costs and the difference cannot be fully paid from the company's treasury.

Clarification: In this case the stock value of the company does not drop before the Recapitalisation Procedure of 6.1 commences.

Once a company becomes insolvent, it goes immediately into Receivership (cf. 6.2), and the following Recapitalisation Procedure 6.1 commences.

6.1 RECAPITALISATION PROCEDURE FOR AN INSOLVENT COMPANY

The following steps are performed in the specified order:

- I. The director and all other shareholders must immediately return all their shares of this company to the bank without compensation. The shares are put into the bank pool. The company's stock value is *not* affected by this action.
- II. If there are still shares of this company in the initial offering, these shares are now transferred to the bank pool, and the company receives the current stock value for each of these shares into its treasury. If there are now enough funds in the treasury to pay the due reorganisation costs (cf. steps VI to IX), skip steps III to V and jump to step VI. If there are still not enough funds, and the company's stock split is already 10%, skip steps III and IV and jump to step V. – Otherwise: If the company has a stock split of 50%, it is now converted to 20% by issuing three new shares. The new shares are put into the bank pool, and the company receives the current stock value for each of these shares into its treasury.
- III. If there are now enough funds in the treasury to pay the due reorganisation costs (cf. steps VI to IX), skip steps IV and V, and jump to step VI. – Otherwise: If the company has a stock split of 20%, it is now converted to 10% by issuing five new shares. The new shares are put into the bank pool, and the company receives the current stock value for each of these shares into its treasury.

- IV. If there are now enough funds in the treasury to pay the due reorganisation costs (cf. steps VI to IX), skip step V and jump to step VI.
- V. If there are still not enough funds in the company's treasury to complete steps VI to IX, then the bank pays the missing money. If this grant is at least as high as the company's current stock value, then the stock value is moved up in the same way as in a profit distribution step as if the grant from the bank were the distributed profit. The bank grants exactly the amount of money that is necessary to perform the following reorganisation steps so that in this case the company has exactly 0 Mark in its treasury after reorganisation.
- VI. If the company is a railway company and became insolvent by case (a), it now buys the next available standard train from the bank.
- VII. If the company became insolvent by case (b), reorganisation is carried out as follows:
- VIII. If the company is a railway company and does not possess a standard train without due maintenance costs in the current game phase, it now buys the next available standard train from the bank. Then all the company's standard trains and switchers with maintenance costs in the current game phase are scrapped.
- IX. If the company is a public mining company, it must replace all obsolete machine markers, i.e. those with maintenance costs in the current game phase. Count the number of obsolete markers and determine how many new machines must be purchased to replace them. It buys exactly as many machines as is necessary to replace the obsolete markers. The replacement is then carried out in the order in which the mines were brought into the company. If a second machine is bought and this one is of higher size than the first one, then extra machine markers are used to successively replace non-obsolete markers of a lower machine size – maybe even ones from the first machine just bought. Then, all the company's switchers with maintenance costs in the current game phase are scrapped.

The company is now recapitalised and reorganised and is thus no longer insolvent. However, the company remains in receivership (cf. 6.2) until it is released from receivership (cf. 6.3).

The company's turn in the current OR is now over.

6.2 COMPANIES IN RECEIVERSHIP

A public company that was formed in the course of the game is in receivership if it is insolvent or no player holds at least 20% of its shares.

As long as a company is in receivership, it operates during its turns in ORs in the following way:

For a public mining company:

- Production and calculation of profit: as usual.
- The profit must be withheld; however, the stock value does *not* drop.

- If due maintenance costs cannot be paid, the company is immediately recapitalised and reorganised as per Rule 6.1.
- The company does *not* buy any new switchers or new mines.
- Except in the case of a recapitalisation and reorganisation as per Rule 6.1, the company neither buys new machines nor does it issue new shares.

For a railway company:

- It builds no new track, neither placing nor upgrading track tiles.
- It does not construct stations.
- It runs its train(s) in the following way: The concession route is run by the company's largest standard train. If this train is at least a 2-train and the company is not the NWE, it runs in addition the route that generates the highest possible revenue, of those routes that are connected to the concession route. If this train is at least a 3-train, the further routes this trains runs are iteratively determined with respect to the highest revenue the respective routes generate. Accordingly, the routes of additional trains are determined, in decreasing size of the trains, always selecting the next possible route with the highest revenue. If more than one route is possible (because of equal revenue), pick one of those randomly. (Note: This procedure does not necessarily give the highest overall revenue.) In addition to the standard train(s), a diesel railcar owned by the company operates as usual.
- Calculation of profit: as usual.
- The profit must be withheld; however, the stock value does *not* drop.
- If due maintenance costs cannot be paid, the company is immediately recapitalised and reorganised (cf. 6.1).
- The company does *not* buy any new switchers.
- Except in the case of a recapitalisation and reorganisation (cf. 6.1), the company neither buys new trains nor does it issue new shares.

6.3 RELEASE FROM RECEIVERSHIP

If a company is in receivership, its shares may not be purchased normally. Instead, there is the following procedure to release the company from receivership:

- If at the start of an auction round one or more companies are in receivership, for each of these companies a purchase option can be auctioned as per Rule 3.1.
- If a player has acquired such a purchase option during an auction round, they may exercise this option in the following stock round by buying 20%–40% of the company's shares (or a 50%-share if the company has a 50% stock split) in a single purchase action as per Rule 3.2.6. After such a purchase, the company is no longer in receivership and the normal rules for buying shares apply.
- Note that if a purchase option acquired during an auction round is not exercised in the following stock round, the option is forfeited and returned to the bank without compensation (cf. 3.2.8). In this case the company remains in receivership.

7. END OF GAME

The game ends as soon as one of the following events occurs:

- The stock of a company hits the highest possible value of 1,000: The game ends after the conclusion of the current OR.
- If, after the start of Phase 5 of the game, the end of the second set of 3 ORs is completed: The game ends immediately.

Every player calculates their wealth: sum the value of each player's cash, shares, and the face values of independent mines owned by the player (not by public mining companies). The player with the greatest wealth wins. Break ties with reverse start turn order.

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9. OVERVIEW OF OPERATING ROUND ACTIONS

INDEPENDENT MINES (4.2)

- Track building (4.2.1) (under certain conditions): up to one tile (pay building cost if applicable).
- Production (4.2.2): calculate revenue (depends on state railway connection, size of the company's machine, and ownership and size of a switcher).
- Profit distribution (4.2.3): pay half of the profit (= revenue minus maintenance costs) to owner, the other half to the company's treasury.
- Purchase machines and/or purchase switchers and/or scrap switchers in any order (4.2.4): machines from the bank only; switchers from the bank or other companies (maintenance costs might be due). Scrapping is always free.
- Closing of the company (4.2.5): Optional. Treasury is paid to the owner.

PUBLIC MINING COMPANIES (4.3)

- Production (4.3.1): calculate revenue (= sum of the revenue of the individual mines).
- Profit distribution (4.3.2): profit (= revenue minus maintenance costs) may be fully distributed, half distributed or fully withheld.
- Stock value adjustment (4.3.3): reduce value one space when profit is withheld; raise value 0 to 3 spaces when profit is distributed, according to the distributed amount.
- Purchase of a mine (4.3.4): up to one mine per OR (open mine from a player or closed mine from the bank – never from other companies). Limit for the maximum number of owned mines applies.
- Purchase machines and/or purchase switchers and/or scrap switchers in any order (4.3.5): machines from the bank only; switchers from the bank or other companies (maintenance costs might be due). Scrapping is always free.
- Issue new shares (4.3.6): only if all existing shares are owned by players.

RAILWAY COMPANIES (4.4)

- Track building (4.4.1): any required number of tiles to completely build the concession route; otherwise lay or upgrade up to 2 tiles (upgrades may limit the number of tiles to 1; pay building costs if applicable).
- Station construction (4.4.2): up to one station per OR; either a new one or an inactive one purchased from another company.
- Running the trains (4.4.3) / Calculation of revenue (4.4.4): revenue from industries (mines, factories), towns and villages of the network serviced by the standard trains of the company (own station markers increase the revenue from towns and villages); a diesel railcar gives additional revenue from towns and villages.
- Profit distribution (4.4.5): profit (= revenue minus maintenance costs) may be fully distributed, half distributed or fully withheld.
- Stock value adjustment (4.4.6): reduce value one space when profit is withheld; raise value 0 to 3 spaces when profit is distributed, according to the distributed amount.
- Purchase trains and/or purchase switchers and/or scrap trains and/or scrap switchers in any order (4.4.7): purchase from the bank or other companies (maintenance costs might be due). Scrapping is always free. The last standard train of a company must not be scrapped.
- Issue new shares (4.4.8): only if all existing shares are owned by players.

MHE (4.5)

- Profit distribution: payment per share is always ten times the size of the MHE's current train (accounts for a total profit of 100 / 200 / 300 / 400 / 500 – depending on the size of the train).
- Purchase one standard train from the bank at no cost if this is the last OR of a set of ORs. Scrap the old train.

10. TILE UPGRADE CHART

Note: yellow tiles are considered to be unlimited in quantity.

TILE	#	UPGRADES	TILE	#	UPGRADES	TILE	#	UPGRADES
	2			2	None			  
	10	  		4	 			  
	4	  		1				 
	4	  		6	  			 
		  		6	  			
				2	  			
	7	  		3	  			
		  						 4 
		  		1	 			 6 
	10	  		1				 5 
		 		1				 2 
		 		1				 2 
		 		1				 2 
		  		1				 2 
		  		1				 1 
				2				 2 None
	2	None		2				 3 None
	2	None		1				 2 None
	1	None			 			 1 None
	2	None						

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